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[Intervention Review]

Interventions for replacing missing teeth: zygomatic implants for the rehabilitation of the severely atrophic edentulous maxilla

Riccardo Federico Visconti^{1a}, Katia Greco¹, Ciro Cotticelli¹, Alessandro Ambrosi², Marco AB Esposito^{1b}

¹Dental School, IRCCS San Raffaele Scientific Institute and Vita-Salute San Raffaele University, Milan, Italy. ²School of Medicine, IRCCS San Raffaele Scientific Institute and Vita-Salute San Raffaele University, Milan, Italy

^aORCID iD: <https://orcid.org/0009-0008-6562-8874>. ^bORCID iD: <https://orcid.org/0009-0004-2952-9933>

Contact: Marco AB Esposito, espositomarco@hotmail.com.

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ABSTRACT

Rationale

Dental implants are widely used to replace missing teeth but require sufficient bone for support. In people with severe maxillary (i.e. upper jaw) bone loss, zygomatic implants provide an alternative to bone augmentation by anchoring in the cheekbone instead of the upper jaw. Potential advantages include avoiding bone grafting and enabling immediate or early placement of a fixed prosthesis. However, the procedure is technically demanding and may be associated with serious complications.

Objectives

To assess the effects of zygomatic implants for the rehabilitation of people with severely atrophic maxillae requiring implant-supported prostheses by comparing: (1) zygomatic implants with or without bone augmentation procedures versus conventional dental implants in augmented bone; (2) zygomatic implants with or without conventional implants versus subperiosteal implants; (3) one type of zygomatic implant or technique versus another type or technique.

Search methods

We used CENTRAL, MEDLINE, Embase, and two trial registers, together with reference checking and contact with study authors to identify the studies included in the review. We conducted searches in November 2025.

Eligibility criteria

We included randomised controlled trials (RCTs), with either a parallel-group or split-mouth design, involving at least 10 participants per arm, with at least one year of follow-up. Participants had severely resorbed maxillae and could not receive conventional dental implants.

Outcomes

Our outcomes were prosthesis and implant failures, complications, patient satisfaction or preference, and cost-effectiveness, assessed as time required to rehabilitate the patient.

Risk of bias

We used Cochrane's RoB 2 tool to assess bias in the RCTs.

Synthesis methods

We did not synthesise data because we included only one study each for two comparisons. We used GRADE to assess the certainty of the evidence.

Included studies

We included two RCTs, each with a three-year follow-up. One parallel-group RCT included 71 participants and compared zygomatic implants—used either alone or in combination with conventional implants and immediately loaded—with delayed placement of conventional implants in bone augmented with bone substitutes (comparison 1). The other split-mouth RCT included 20 participants and compared the placement of zygomatic implants using conventional rotary drills with piezoelectric preparation on the contralateral side (comparison 3).

Synthesis of results

Comparison 1: zygomatic implants with and without bone augmenting procedures versus conventional dental implants in augmented bone

After three years, zygomatic implants may reduce prosthetic failures compared with conventional dental implants placed in grafted bone (odds ratio (OR) 0.29, 95% confidence interval (CI) 0.05 to 1.56; 66 participants; low-certainty evidence). Zygomatic implants probably reduce implant failures compared with conventional implants (OR 0.23, 95% CI 0.06 to 0.96; 61 participants; moderate-certainty evidence). Zygomatic implants probably increase complications (OR 4.53, 95% CI 1.47 to 13.98; 66 participants; moderate-certainty evidence). Zygomatic implants may have little to no effect on patient satisfaction compared with conventional implants in grafted bone (MD 0.40, 95% CI -3.05 to 3.85; 60 participants; low-certainty evidence). As a measure of cost-effectiveness, time to rehabilitation is probably shorter with zygomatic implants compared with conventional implants in grafted bone (MD -442.98 days, 95% CI -518.64 to -367.32; 64 participants; moderate-certainty evidence), corresponding to approximately one day (1.3 days) with zygomatic implants versus more than one year (444 days) with bone grafting.

Comparison 3: different zygomatic implants and techniques: piezoelectric surgery compared with rotary drills

After three years, preparation with piezoelectric surgery may have little to no effect on prosthetic failures compared with rotary drills, and the evidence is compatible with benefit, no effect, and harm (OR 1.00, 95% CI 0.02 to 50.40; 17 participants; low-certainty evidence). Piezoelectric surgery may have little to no effect on implant failures compared with rotary drills, with very wide confidence intervals compatible with benefit, no effect, and harm (OR 1.00, 95% CI 0.02 to 50.40; 17 participants; low-certainty evidence). Complications may be similar between rotary drills and piezoelectric surgery (OR 1.50, 95% CI 0.25 to 8.99; 20 participants; low-certainty evidence). Patient preference may favour piezoelectric surgery, but the estimate is very imprecise and compatible with both a strong preference for piezoelectric surgery and equal preference for the two approaches (OR 9.00, 95% CI 0.48 to 167.26; 20 participants; low-certainty evidence). As a measure of cost-effectiveness, implant placement is probably faster with rotary drills than with piezoelectric surgery (MD -9.15 minutes, 95% CI -10.41 to -7.89; 20 participants; moderate-certainty evidence), corresponding to approximately 14 minutes with rotary drills versus 24 minutes with piezoelectric surgery.

The certainty of the evidence for both comparisons ranged from moderate to low. We downgraded mainly for serious imprecision resulting from small sample sizes and wide confidence intervals. The risk of bias was generally low, although we judged one outcome (patient satisfaction) in one study to be at high risk of bias due to missing outcome data.

Authors' conclusions

At three years, zygomatic implants probably reduce time to rehabilitation compared with bone grafting (1.34 versus 444.32 days). They probably reduce implant failures and probably increase the risk of complications. Zygomatic implants may reduce prosthetic failures, but the estimates are imprecise. The effect on patient satisfaction remains uncertain.

Compared with piezoelectric surgery, rotary drills probably reduce implant placement time. For prosthetic failures, implant failures, complications, and patient preference, piezoelectric surgery may have little to no effect, and important benefit or harm cannot be ruled out because of very wide confidence intervals. Implant site preparation with piezoelectric surgery is probably slower and may not always be feasible in the presence of very dense bone.

Overall, the evidence base for both comparisons is limited and is affected by imprecision, which restricts the confidence with which these findings can be applied in clinical settings. More well-designed RCTs are needed to reduce the current uncertainty.

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Registration

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PLAIN LANGUAGE SUMMARY

What are the benefits and risks of cheekbone dental implants for people with no teeth and severe bone loss in the upper jaw?

Key messages

- Compared to standard dental screw implants placed after building up the jawbone, longer screws placed in the cheekbone were less likely to fail and allowed working replacement teeth to be fitted much sooner, but led to more complications.
- Drilling the holes that receive the implant screws can be done with a standard rotary (spinning) tool or a vibrating tool that 'shaves' the bone (piezoelectric surgery). There may be little or no difference between the two methods in terms of how durable the implants or replacement teeth are, or the risk of complications, but rotary drills are probably faster.
- Few studies have looked at these issues. We need more and bigger studies to have more confidence about the benefits and risks.

What happens when people have no or few natural teeth?

When teeth are lost, the surrounding bone can gradually shrink because it is no longer stimulated by chewing. The amount of bone loss varies from person to person. It can be so severe that people are unable to wear removable dentures comfortably. This can make eating and speaking difficult and may affect confidence, well-being, and social relationships. Dental implants can help address these problems.

What are dental implants, and how do they help?

Standard dental implants are short screws (0.5 to 2 cm long) placed into the jawbone which act like artificial tooth roots. Dentists can attach permanent ('fixed') teeth to them, or removable or semi-removable dentures.

Cheekbone implants ('zygomatic implants') are longer: about 3 to 5 cm. They are used when the upper jawbone has shrunk so much that there is not enough bone left to hold standard implants securely. Cheekbone implants almost always support fixed teeth, not removable dentures.

With cheekbone implants, fixed teeth can often be placed on the same day as the implants. This is because they avoid the need for jawbone grafting, which is when a surgeon adds new bone (or bone-like material) to the jaw so it can support an implant. Bone grafting is complex, expensive, and time-consuming. It may require several surgeries and take more than a year before replacement teeth can be fitted.

Cheekbone implant screws can be placed using rotary drills or piezoelectric surgery, which uses sound vibrations to cut bone and may be less traumatic. Cheekbone dental implants can lead to complications, and the potential benefits need to be weighed against these risks.

What did we want to find out?

We wanted to find out if, compared to standard implants in grafted jawbone, cheekbone implants led to fewer:

- replacement teeth failures – where the teeth break or loosen;
- implant failures – where the implant screws become unstable;
- complications.

We also wanted to find out:

- which type of implant led to higher satisfaction and faster rehabilitation;
- which method for implant site preparation was faster for surgeons and preferred by people undergoing treatment.

What did we do?

We searched for studies that compared:

- cheekbone implants to standard implants in grafted jawbone in people with severe bone loss in the upper jaw; or
- different ways to prepare the cheekbone implant site.

We compared and summarised the study results, and rated our confidence in the evidence, based on factors such as study methods.

What did we find?

We found two small studies that followed participants for three years. They were both carried out in Europe. One compared cheekbone implants to standard implants in grafted bone, and involved 71 people. The other compared piezoelectric surgery to rotary drills for preparing the implant site, and involved 20 people.

Main results

After three years, compared to standard implants in grafted jawbone, cheekbone implants:

- may have led to fewer replacement teeth failures: for every 100 people treated, about 12 fewer people with cheekbone implants had problems with their replacement teeth;
- probably had fewer implant failures: for every 100 people treated, about 22 fewer people with cheekbone implants had problems;
- probably led to more complications: roughly four-fifths of people with cheekbone implants had complications, compared to about half of the people with standard implants.

We do not know which procedure left people more satisfied. However, people who received cheekbone implants probably received their new teeth much sooner – in a little more than 1 day – than those with standard implants, where it typically took 444 days before replacement teeth could be fitted.

In the study that compared two methods (piezoelectric surgery versus rotary drills) for preparing the implant sites, after three years, there may be little or no difference between them in terms of:

- replacement teeth failures;
- implant failures;
- complications;
- people's preference.

However, rotary drills probably made implant placement faster by more than 9 minutes.

What are the limitations of the evidence?

Our confidence in the evidence ranged from very low to moderate. We found only two studies testing these approaches, and they involved few people. More well-designed studies are needed to identify the best methods.

How up to date is this evidence?

The evidence is current to November 2025.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Zygomatic implants compared to conventional dental implants in grafted bone for people with severely atrophic maxillae

Zygomatic implants compared to conventional dental implants in grafted bone for people with severely atrophic maxillae

Patient or population: people with severely atrophic maxillae

Setting: hospital and dental practice

Intervention: zygomatic implants

Comparison: conventional dental implants in grafted bone

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with conventional dental implants in grafted bone	Risk with zygomatic implants				
Prosthetic failures follow-up: 3 years	182 per 1000	61 per 1000 (11 to 257)	OR 0.29 (0.05 to 1.56)	66 (1 RCT)	⊕⊕⊕⊕ Low ^a	
Implant failures follow-up: 3 years	310 per 1000	94 per 1000 (26 to 302)	OR 0.23 (0.06 to 0.96)	61 (1 RCT)	⊕⊕⊕⊕ Moderate ^b	
Complications follow-up: 3 years	485 per 1000	810 per 1000 (580 to 929)	OR 4.53 (1.47 to 13.98)	66 (1 RCT)	⊕⊕⊕⊕ Moderate ^b	Complications experienced in zygomatic group: sinus epithelium perforation, postoperative swelling, infraorbital nerve paraesthesia, implant failure, pain, sinusitis, facial fistula, peri-implant mucosa recession, peri-implantitis, mucositis, implant exposure, prosthesis fracture, abutment screw loosening or fracture, tooth detachment. Complications experienced in grafted bone group: sinus or nasal perforation, graft infection, peri-implantitis, implant exposure or migration.
Patient satisfaction assessed with: Oral Health Impact Profile-14; lower scores indicate better quality of life Scale from: 0 to 56	The mean patient satisfaction was 4.11	MD 0.4 higher (3.05 lower to 3.85 higher)	-	60 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,c}	

follow-up: 3 years					
Time to rehabilitate the patient (cost-effective-ness) assessed with: days	The mean time to rehabilitate the patient was 444.32 days	MD 442.98 days lower (518.64 lower to 367.32 lower)	-	64 (1 RCT)	⊕⊕⊕⊖ Moderate ^d

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; MD: mean difference; OR: odds ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_462718533010298051.

^a Downgraded by two levels for very serious imprecision because the confidence interval was very wide and included the possibility of no effect

^b Downgraded by one level for imprecision due to the wide confidence interval

^c Downgraded by one level for risk of bias (bias due to missing outcome data)

^d Downgraded by one level for imprecision due to small numbers of participants

Summary of findings 2. Piezoelectric surgery versus rotary drills for zygomatic implant site preparation in people with severely atrophic maxillae

Patient or population: people requiring dental implants

Settings: dental practice

Intervention: piezoelectric surgery preparation

Comparison: rotary drills

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	No. of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with rotary drills	Risk with piezo-electric surgery preparation				

Prosthetic failures (3 years)	59 per 1000	59 per 1000 (1 to 760)	OR 1.00 (0.02 to 50.40)	17 (1 RCT)	⊕⊕○○ Low ^a	One prosthetic failure occurred in 1 of 17 participants and affected both the piezoelectric and rotary sides.
Implant failures (3 years)	59 per 1000	59 per 1000 (1 to 760)	OR 1.00 (0.02 to 50.40)	17 (1 RCT)	⊕⊕○○ Low ^a	One implant failure occurred in 1 of 17 participants and affected both the piezoelectric and rotary sides.
Complications (3 years)	300 per 1000	391 per 1000 (97 to 794)	OR 1.50 (0.25 to 8.99)	20 (1 RCT)	⊕⊕○○ Low ^a	Drills: inferior lip burns, zygoma fracture, bilateral infection, peri-implant mucositis, sinusitis, prosthesis screw fracture Piezosurgery: bilateral infection, implant exposure, implant extruding through the zygoma, prosthesis screw fracture, peri-implant mucositis
Patient preference (3 years)	0 per 1000	0 per 1000 (0 to 0)	OR 9.00 (0.48 to 167.26)	20 (1 RCT)	⊕⊕○○ Low ^a	Four of 20 participants preferred the piezoelectric surgery side, whereas no participants preferred the rotary side. Sixteen participants reported no preference.
Implant placement time (cost-effectiveness) assessed with: minutes	The mean implant placement time was 14.35 minutes	MD 9.15 minutes longer (7.89 longer to 10.41 longer)	-	20 (1 RCT)	⊕⊕⊕○ Moderate ^b	

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; OR: odds ratio

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

^aDowngraded by two levels for imprecision because all confidence intervals were extremely wide and included the possibility of benefit, no effect, and harm

^bDowngraded by one level for imprecision due to small sample size

BACKGROUND

Description of the condition

Dental implants are used to replace missing teeth by anchoring artificial 'roots' into the jawbone. However, placing dental implants can be challenging when there is insufficient bone volume. People who have no teeth (known as 'edentulous'), whose jawbone has atrophied (known as 'resorbed': where bone has gradually shrunk or thinned after tooth loss), or who have lost bone through traumatic injury or resective surgery for oral cancer may all have insufficient bone volume for dental implant placement. Therefore, these people may suffer from unstable dentures due to lack of retention. For this reason, they struggle with both eating and speaking, which often leads to depression and social alienation [1, 2].

Several bone augmentation procedures have been developed to overcome insufficient bone volumes. Onlay and inlay bone grafting, as well as sinus or nasal floor augmentation, are among the most frequently employed techniques. However, for major bone grafting, surgeons often need to take bone from another part of the person's body. These larger procedures often mean higher morbidity, and require general anaesthesia and several days in the hospital for recovery. Moreover, treatment costs are substantial, and the overall rehabilitation process may extend beyond one year before a fixed prosthesis can be attached to the implants [3, 4]. From a health-economic perspective, prolonged rehabilitation times imply multiple surgical interventions, additional clinical appointments, delayed functional recovery, and extended periods without a fixed prosthesis, all of which contribute to increased direct and indirect costs. During this time, people may remain without a prosthesis, experiencing difficulties with speech, mastication (i.e. chewing), and social interactions.

Description of the intervention and how it might work

In recent decades, for the rehabilitation of severely deficient edentulous maxilla, several alternatives to bone augmentation procedures have been proposed, including implant placement in the pterygomaxillary region [5], the use of short and/or tilted implants [6, 7], trans-sinus implants [8], nasal implants [9], subperiosteal implants [10], and zygomatic implants [11].

Introduced by Professor P-I Brånemark in the late 1980s [11], zygomatic implants are long, screw-shaped implants placed through the maxillary bone to engage the zygomatic bone. Their main indication is the rehabilitation of people without teeth and with extreme maxillary bone atrophy [11, 12].

In Brånemark's original protocol [11], the implant was inserted from the alveolar crest, entirely traversing the maxillary sinus. A lateral window approximately 10 mm wide was created in the sinus wall to visualise the correct trajectory and ensure engagement of the zygomatic bone. To minimise the risk of orbital perforation, implants were often placed with a pronounced palatal inclination, which resulted in an excessively palatal emergence of the implant head. This configuration constrained tongue space, compromised prosthetic design, and complicated oral hygiene. Sinusitis and other sinus-related complications were relatively common. To mitigate these issues, new surgical techniques were developed to allow an extra-sinus trajectory for zygomatic implants, taking into account individual anatomic variations [13, 14, 15, 16]. The

slot technique, proposed by Stella [14], involved creating a slot in the vestibular bony wall of the maxilla, allowing the implant to be positioned more externally. This modification brought the implant head closer to the natural tooth position, improving the prosthetic profile and patient comfort; however, sinusitis remained relatively frequent. To further reduce the risk of chronic sinusitis, later approaches positioned the implant almost completely extra-sinus [15], with the head resting on rather than engaging the crestal bone. This technique, while decreasing sinus involvement, led to a new complication—soft-tissue retraction—resulting in exposure of the coronal implant surface within the oral cavity and chronic peri-implant mucosal inflammation. The optimal implant trajectory and emergence profile therefore remain subjects of ongoing debate.

Between one and three zygomatic implants may be placed on each side of the maxilla. In cases with residual anterior bone, two zygomatic implants (one per side) can be combined with two conventional anterior implants or with pterygoid implants in the posterior region. In people with severely atrophic maxillae, however, the most common rehabilitation involves two (occasionally three) zygomatic implants per side supporting a fixed prosthesis [17, 18]. Recently, there is a tendency by surgeons to add one pterygoid implant per side in addition to four zygomatic implants.

The procedure requires advanced surgical expertise and may be performed under general anaesthesia, conscious sedation, or local anaesthesia, although the latter is generally less advisable. Importantly, due to the potential for serious complications that can be difficult to manage, zygomatic implants should be reserved for cases of extreme maxillary atrophy, where placement of four conventional implants is not feasible.

The main advantages of zygomatic implants are the avoidance of complex bone-augmentation procedures and the possibility of immediate loading when adequate insertion torque is achieved, thus enabling delivery of a fixed prosthesis on the same day [18, 19, 20, 21].

The reported complications of zygomatic implants include:

- intraoperative complications, such as periorbital cavity penetration [18, 22], zygoma fracture [23], and subcutaneous malar emphysema [24];
- postoperative complications, such as moderate nasal bleeding for one to three days [25], facial or periorbital haematoma [25], temporary and permanent sensory nerve deficits [25], intraoral soft tissue infection [18], sinusitis [21], and oroantral fistulas [26];
- long-term complications, such as gingival hyperplasia [27], peri-implantitis [28], sinusitis [21], oroantral fistulas [26], cutaneous fistulas [29], zygomatic implant failures [11], and implant fenestration through the soft tissue [30].

There is broad consensus that zygomatic implants were originally conceived for, and should mainly be reserved for, the rehabilitation of severely atrophic maxillae. This was highlighted in recent consensus reports stating that “zygomatic implants should be a last-resort solution when conventional treatments are not feasible” [31, 32].

Although rehabilitation via four conventional implants represents a potential treatment option for some people with atrophic maxillae,

these approaches are not always possible in cases of severe maxillary atrophy. In such clinical scenarios, zygomatic implants may represent an alternative to bone augmentation procedures aimed at allowing the placement of conventional implants.

The main indications for zygomatic implants are:

- treatment of severely atrophic fully edentulous maxillae without using any bone augmentation procedure;
- treatment of severely atrophic partially edentulous maxillae, avoiding bone augmentation procedures (grafting or sinus lifting);
- maxillary reconstruction in people affected by maxillectomy defects. Zygomatic implants may fix maxillary obturators as an alternative to non-implant retained obturators, microvascular free flaps, and local regional flaps [11, 33, 34].

Before deciding to opt for zygomatic implants, clinical and radiographic examinations are needed. Three-dimensional computed tomography is essential for treatment planning, to assess the individual anatomical characteristics and to plan the angulation and the emergence site of the implants [35].

Recently, subperiosteal implants—originally introduced in the mid-1900s—have regained interest owing to advances in digital imaging and three-dimensional printing technologies. These implants consist of custom-designed metal frameworks that are positioned on the bone surface beneath the periosteum and secured with screws, offering a potential treatment option for people with severe bone atrophy. They are immediately rehabilitated with a fixed prosthesis. This approach has attracted increasing clinical attention, as it appears to be simpler and less invasive than zygomatic implant placement [10]. The growing interest in subperiosteal implants as an alternative to zygomatic implants in the rehabilitation of severely resorbed maxillae is supported by a 2025 systematic review on this topic, which included only non-randomised studies [36].

Over recent years, numerous innovations have been introduced to optimise the use of zygomatic implants, as demonstrated by four recently published systematic reviews [37, 38, 39, 40]. Advances in digital imaging and computer-assisted planning have enabled the development of navigated and guided surgery systems, which employ digitally designed surgical guides or real-time navigation to assist zygomatic implant placement, as an alternative to purely visual guidance. These technologies are intended to improve accuracy, reduce intraoperative risks, and standardise the procedure compared with the conventional free-hand technique. However, the potential advantages and risks of these methods remain uncertain, as robust clinical evidence is still lacking [37, 38, 39, 40].

Another unresolved issue concerns the optimal trajectory for zygomatic implant placement—whether intra-sinus, within a sinus slot on the vestibular aspect of the maxilla, or extra-sinus.

The placement of such long implants requires the use of extended and rigid drills, which are difficult to cool adequately with irrigation and may cause thermal injury or mechanical trauma due to vibration and deflection. Piezoelectric devices have been proposed as a less traumatic alternative, offering selective cutting of mineralised tissues while preserving adjacent soft structures.

Nonetheless, piezosurgery may have limited cutting efficiency in very dense zygomatic bone and may prolong surgical time.

Additional technical considerations relate to the diameter of zygomatic implants, which must be sufficient to withstand masticatory loads and transmit forces effectively to the zygomatic bone, yet narrow enough to allow placement in people with smaller zygomas and to minimise the risk of bone fracture.

Why it is important to do this review

People with extremely resorbed jaws are often considered functionally compromised, as they are unable to wear conventional dentures. Consequently, they experience significant difficulties with eating and speaking, which can markedly reduce their quality of life and may lead to social withdrawal or psychological distress. The rehabilitation of these individuals with fixed prostheses is complex, lengthy, costly, and often associated with discomfort, as it frequently requires multiple bone grafting procedures to reconstruct sufficient bone volume for implant placement. Furthermore, many people may not have the financial resources or the physical resilience to undergo these extensive procedures.

Alternative, less invasive interventions, such as zygomatic or subperiosteal implants, have been proposed to overcome the limitations of conventional bone augmentation, potentially reducing treatment time, costs, and patient morbidity. However, several systematic reviews have emphasised both the frequency and the severity of complications that may be associated with zygomatic implants [22, 41, 42, 43, 44, 45]. In particular, sinusitis with a reported prevalence of up to 14.2% [41], and accidental damages to the orbit with related ophthalmological complications [22], have been described. To reduce the risk of sinusitis, prophylactic meatal antrostomy has been proposed [24, 46, 47].

Despite the possible occurrence of complications, the rehabilitation of people with severely atrophic maxillae using zygomatic implants or bone augmentation and conventional implants appears to substantially improve individuals' quality of life [48]. Furthermore, a systematic review of patient-reported outcomes in individuals with severe maxillary atrophy rehabilitated with zygomatic implant-supported prostheses reported marked improvements in oral health-related quality of life and high levels of overall satisfaction, particularly with regard to chewing ability and aesthetics [49]. However, although successful rehabilitation procedures are generally associated with significant improvements in quality of life, it remains unclear whether any specific intervention leads to a greater degree of improvement than others.

This is an update of a Cochrane review first published in 2003 [50], and previously updated in 2005 and 2013 [51, 52], none of which identified any eligible randomised controlled trials. Since 2013, new clinical evidence has become available, warranting a reassessment of the comparative effectiveness and safety of these alternative techniques.

The purpose of this updated review is therefore to evaluate the current evidence on zygomatic implants for restoring function in people with severely atrophic maxillae, and in particular whether they offer a safer, more effective, and more cost-efficient alternative to traditional bone augmentation procedures for placing standard implants.

OBJECTIVES

To assess the effects of zygomatic implants for the rehabilitation of people with severely atrophic maxillae requiring implant-supported prostheses by comparing: (1) zygomatic implants with or without bone augmentation procedures versus conventional dental implants in augmented bone; (2) zygomatic implants with or without conventional implants versus subperiosteal implants; (3) one type of zygomatic implant or technique versus another type or technique.

METHODS

We followed guidance in the Methodological Expectations for Cochrane Intervention Reviews (MECIR) manual when conducting the review [53], and PRISMA 2020 for the reporting [54].

Differences between protocol and review

For a description of differences between the protocol and previous versions of this review that we carried forward in this update, please see [Supplementary material 8](#) [3, 50, 51, 52, 55].

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs) with either a parallel-group or split-mouth design. Eligible studies enrolled people with severely resorbed maxillae who could not receive conventional dental implants, included at least 10 participants per study arm for parallel-group trials (or at least 10 participants in split-mouth trials), and reported outcomes after a minimum follow-up of one year.

Types of participants

We included participants with a severely atrophic maxilla who could not be rehabilitated with conventional dental implants due to lack of bone, including people subjected to cancer ablation surgery and irradiation therapy. Both partially and completely edentulous participants were eligible.

Types of interventions

We included studies that compared:

- the efficacy of zygomatic implants with and without bone augmenting procedures in comparison with conventional dental implants in augmented bone;
- the efficacy of zygomatic implants with and without conventional implants in comparison with subperiosteal implants;
- different zygomatic implants and techniques.

Outcome measures

We planned to present data stratified by follow-up duration. For studies reporting multiple follow-up time points, we extracted and analysed results at the longest available follow-up within the following time frames:

- short-term follow-up (>1 year to ≤ 3 years);
- medium-term follow-up (> 3 years to < 10 years);
- long-term follow-up (≥ 10 years).

Critical outcomes

- **Prosthetic failures.** We defined this outcome as a prosthesis that could not be placed, or the loss of a prosthesis secondary to implant failure or replaced for any other reason.
- **Implant failures.** We defined this outcome as mobile implants, removal of stable implants dictated by progressive marginal bone loss or infection, implant fractures, or implants left sleeping (i.e. implants that are placed but not loaded or connected to the prosthesis) due to improper alignment. Failures were reported at participant level for parallel-group studies and at side level for split-mouth studies.
- **Complications.** We defined this outcome as any biological or technical adverse event occurring during or after surgery or over the entire follow-up, including sinusitis, soft tissue dehiscence, mucosal inflammation, prosthetic screw loosening or fracture, and any other complications as reported by the investigators. Complications were reported at participant level for parallel-group studies and at sides of the mouth for split-mouth studies.
- **Patient satisfaction.** We defined this outcome to include patient-reported quality of life (assessed through validated instruments, questionnaires, or visual analogue scales) and, when possible, patient preference in split-mouth studies.

Important outcomes

- **Cost-effectiveness.** We defined this outcome as measures reflecting the efficiency of the intervention, including the total time required to rehabilitate the patient, the number of surgeries or procedures performed, and the duration of implant placement.

Search methods for identification of studies

All search methods and search strategies are presented in [Supplementary material 1](#). This update includes searches up to 11 November 2025.

Electronic searches

We searched the following electronic databases:

- Cochrane Central Register of Controlled Trials (CENTRAL; 2025, Issue 11) in the Cochrane Library (searched from 1 January 2013 to 11 November 2025);
- MEDLINE via OVID (1 January 2013 to 11 November 2025);
- Embase via OVID (1 January 2013 to 11 November 2025).

The search strategies were based on strategies developed by Cochrane Oral Health's Information Specialist for the previous version of this review [52]. We modified these searches to include more specific terms for implants in zygomatic bone. We combined subject-specific terms with Cochrane's highly sensitive search strategy for identifying randomised trials in MEDLINE and Embase [56]. In this review update, we searched all databases from 2013 to 11 November 2025, with no restriction on language of publication.

We also updated our searches of the World Health Organization (WHO) International Clinical Trials Registry Platform (ICTRP) and Clinicaltrials.gov portals for ongoing and unpublished trials (to 19 November 2025).

Searching other resources

We wrote to all the authors of any ongoing RCTs to identify unpublished studies. We checked the bibliographies of all identified RCTs and relevant review articles. We also used personal contacts to try to identify unpublished or ongoing RCTs.

To determine if any eligible studies had been retracted, we ran searches in Retraction Watch using reference information from eligible studies.

Data collection and analysis

Review authors with direct involvement in the conduct of any eligible study were not involved in study selection, data extraction, risk of bias assessment, or GRADE assessment.

Selection of studies

Two review authors (RFV, CC) independently screened titles and abstracts identified in the search using Covidence [57]. For studies appearing to meet the inclusion criteria, or for which there were insufficient data in the title and abstract to make a clear decision, we obtained the full report. The same two review authors independently assessed the reports to establish whether the studies met the inclusion criteria. Had there been any disagreement between review authors, or if a study authored by one of the review authors was considered for inclusion, a third review author (KG) acted in place of that author. In these instances, the study author was not involved in the selection process; the remaining two review authors independently conducted study selection. All studies meeting the inclusion criteria underwent risk of bias assessment and data extraction. We recorded any studies rejected at this or subsequent stages in the table of excluded studies, along with reasons for exclusion.

Data extraction and management

Two review authors (RFV, CC) extracted data, independently and in duplicate, using specially designed data extraction forms. If a study authored by one of the review authors was considered for inclusion, a third review author (KG) acted in place of that author. In these instances, the study author was not involved in the data extraction process; the remaining two review authors independently extracted data. We consulted another review author (KG) in case of disagreements over data extraction.

For each included study, we recorded the following data:

- publication year, country of origin, setting, and source of study funding;
- participants' demographic characteristics and the criteria for inclusion;
- details about the type of intervention;
- details about the outcomes reported, including assessment method and assessment time frames.

Risk of bias assessment in included studies

Two review authors (RFV, CC) independently assessed risk of bias using RoB 2 [58, 59, 60], considering the effect of assignment to intervention (intention-to-treat effect). If a review author was also the author of an included study, two other review authors who were not involved in the study independently evaluated the report.

We assessed risk of bias for all RoB 2 domains and judged each domain as having 'high risk of bias', 'some concerns', or 'low risk of bias' using the responses to the signalling questions and algorithms within the RoB 2 tool. We used the tool algorithm to reach an overall risk of bias assessment for each outcome. We quoted evidence to support our judgements and if we disagreed with a judgement recommended by the algorithm, we included an explicit explanatory statement [60]. We assessed the risk of bias for all critical and important outcomes at the longest available follow-up.

Measures of treatment effect

In both parallel-group and split-mouth studies:

- for dichotomous outcomes, we expressed the estimate of the effect of the intervention as an odds ratio (OR) together with its 95% confidence interval (CI);
- for continuous outcomes, we used mean differences (MDs) and 95% confidence intervals to summarise the data for each study.

Unit of analysis issues

We analysed data using participants as the unit of analysis, apart from split-mouth studies, in which the sides of the mouth were considered as the units of analysis.

For split-mouth studies, we analysed dichotomous outcomes using paired odds ratios based on discordant pairs only, by comparing the number of participants who experienced the event with one intervention but not with the other to those who experienced the event with the other intervention but not with the first.

Had studies with more than two intervention groups been eligible, we would have incorporated them in the review following the guidance in the *Cochrane Handbook for Systematic Reviews of Interventions* (Section 6.2.9) [61]. Specifically, we would have avoided double-counting of participants by either combining relevant intervention groups where clinically appropriate or by splitting the shared comparator group across pair-wise comparisons, ensuring that each participant contributed only once to each meta-analysis and that unit-of-analysis errors were avoided.

Dealing with missing data

Where necessary, we planned to contact study authors to request clarification about study details, obtain missing data, or both. If we were unsuccessful in obtaining these data, we planned to exclude the study.

Reporting bias assessment

If we had included at least 10 studies, we planned to generate a funnel plot to explore potential small-study and publication biases.

Synthesis methods

We conducted statistical syntheses using Review Manager (RevMan) software [62]. We planned to assess statistical heterogeneity using the I^2 statistic and to use a fixed-effect model as the primary approach because of the small number of included studies and the clinical homogeneity of the interventions [63].

For parallel-group studies with dichotomous outcomes, we planned to pool treatment effects as odds ratios (ORs) with

95% confidence intervals using the inverse-variance method. For continuous outcomes measured on the same scale, we planned to calculate pooled estimates as mean differences (MDs) with 95% confidence intervals.

For split-mouth studies, we analysed dichotomous outcomes as paired binary data. We extracted individual paired data directly from the study and calculated paired odds ratios based exclusively on discordant pairs. We compared the number of participants who experienced the event with one intervention but not with the other with the number of participants who experienced the event with the alternative intervention but not with the first. When one discordant cell contained zero events, we applied a continuity correction of 0.5 [58]. We calculated the variance of the log odds ratio as the sum of the inverse of the corrected discordant cells and entered the paired effect estimate and its standard error into RevMan using the generic inverse-variance method, in accordance with methods described by Elbourne and colleagues [64]. If odds ratios were available, we planned to extract them directly from the study. For continuous outcomes in split-mouth designs, we used paired mean differences when available and extracted them from the study.

We planned to combine data from split-mouth and parallel-group studies using the methods described by Elbourne and colleagues [64].

We intended to perform meta-analyses where studies were sufficiently homogeneous in terms of participants, interventions, and outcome measures with comparable follow-ups. However, we could not conduct a quantitative synthesis because we included only two studies with different aims.

Investigation of heterogeneity and subgroup analysis

We planned to assess clinical heterogeneity by examining the types of participants, interventions, and outcomes in each study, with a planned subgroup analysis according to different bone grafting techniques and participant groups (patients with severely resorbed maxillae versus cancer patients who had undergone irradiation therapy).

Sensitivity analysis

We planned to assess the impact of excluding studies with outcomes judged to be at high risk of bias in any of the RoB 2 domains (randomisation process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result) from the meta-

analyses. We also planned to assess the impact of including unpublished literature in the meta-analyses. However, because meta-analysis was not feasible, we did not perform these sensitivity analyses.

Certainty of the evidence assessment

We created a summary of findings table for each prespecified comparison, presenting all five critical and important outcomes at the longest follow-up where data were available.

We used the five GRADE considerations (overall risk of bias, consistency of effect, imprecision, indirectness, and publication bias) to assess the certainty of the evidence.

We used the methods and recommendations described in Chapter 14 of the *Cochrane Handbook for Systematic Reviews of Interventions* to apply the GRADE approach, using GRADEpro GDT software [65, 66].

Equity considerations

We did not investigate equity-related characteristics in this review.

Consumer involvement

Consumers were not involved in this review.

RESULTS

Description of studies

See [Supplementary material 2](#), [Supplementary material 3](#), and [Supplementary material 4](#).

A complete record of all comparisons and analyses is available for this review in [Supplementary material 6](#) and a full data package is available in [Supplementary material 7](#).

Results of the search

The searches for this update covered June 2013 to 11 November 2025. We identified a total of 469 records: 466 through database searching and three through other sources. After deduplication, we screened 402 records by title and abstract, and excluded 375 as irrelevant to the review. We retrieved 27 full-text articles for detailed eligibility assessment. We included two studies (five references), identified four ongoing studies (four references), and excluded 17 studies (18 references). [Figure 1](#) shows the screening process for this update.

Figure 1. PRISMA flow diagram of study selection for the review update

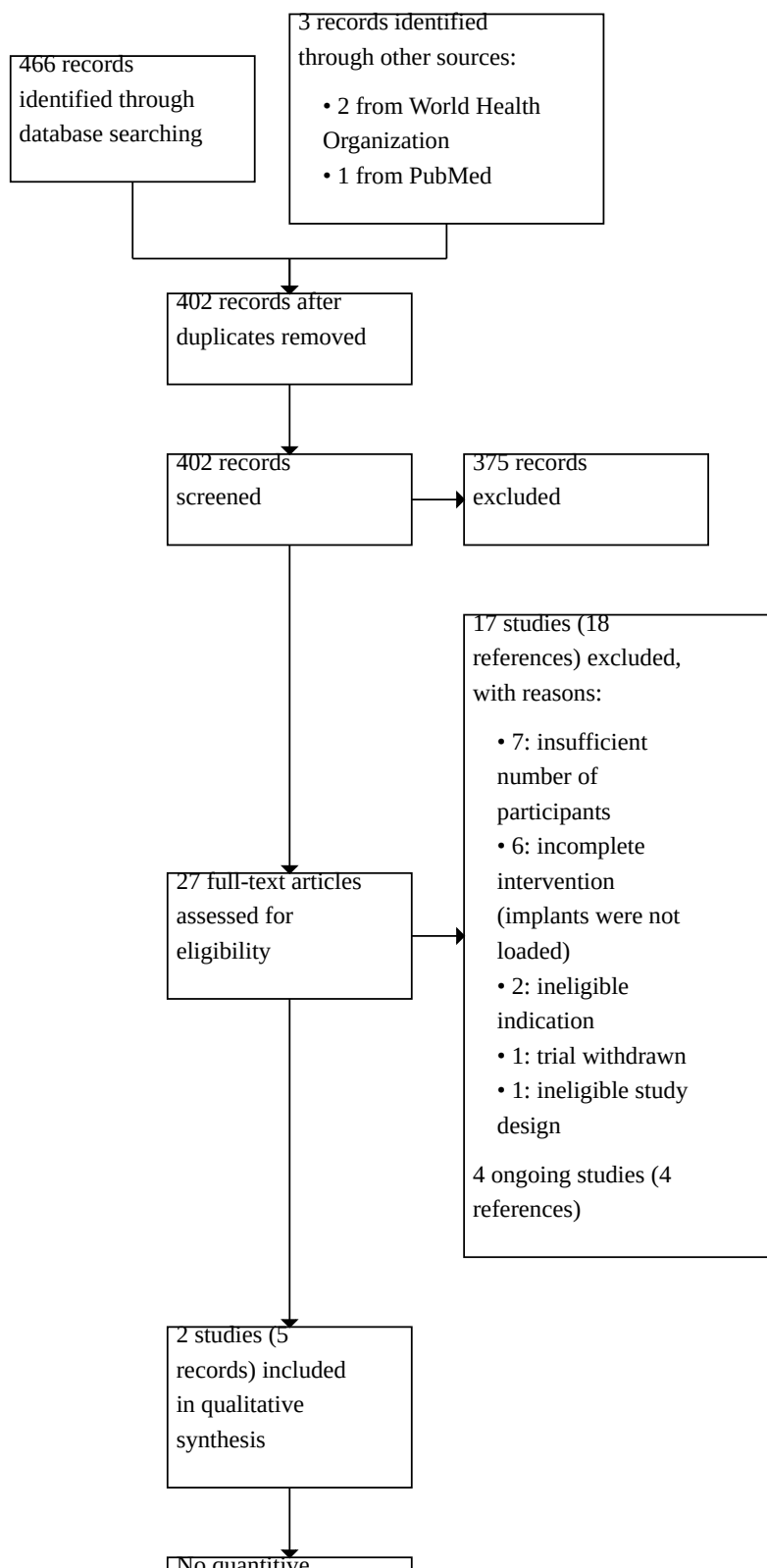
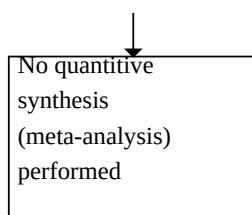


Figure 1. (Continued)



Included studies

We included two studies (Esposito 2017 [67, 68]; Esposito 2018 [48, 69, 70]). [Table 1](#) summarises the included studies. [Supplementary material 2](#) provides key study characteristics important for interpreting the syntheses.

Study characteristics

The included studies were conducted in Europe (Spain and Italy) and published in 2017 and 2018. Esposito 2018 was a multicentre parallel-group RCT with a 3-year follow-up that compared immediately loaded zygomatic implants with conventional implants placed in bone augmented with a bone substitute. Participants were treated across three centres: Hospital Clinic, Barcelona (Spain); Policlinico Sant'Orsola Malpighi, Bologna (Italy); and Ospedale San Filippo Neri, Rome (Italy). Esposito 2017 was a split-mouth RCT, conducted at a single centre by two independent external teams with a 3-year follow-up. It compared implant site preparation using piezosurgery versus rotary instruments. All procedures were performed at the University Hospital of Ancona (Italy).

Participant characteristics

Both studies exclusively included adult participants (≥ 18 years) with edentulous and severely atrophic maxillae unsuitable for conventional implant placement without advanced surgical procedures. Inclusion criteria for each study are described in [Supplementary material 2](#).

Treatment format

In Esposito 2018, treatment consisted either of an immediate full-arch rehabilitation supported by four zygomatic implants or by two zygomatic and two anterior conventional implants, immediately loaded within one week, or of a staged reconstructive approach based on bone augmentation followed by placement of six to eight conventional implants and delayed loading. In Esposito 2017, two different techniques for zygomatic implant site preparation were compared within the same participants using a split-mouth design, with piezoelectric surgery on one side and conventional rotary drills on the contralateral side; the four implants on both sides were immediately loaded under the same prosthesis.

Outcome reporting and assumptions required for synthesis

Both trials reported failure outcomes (prosthetic and implant failures), safety outcomes (complications), patient-centred outcomes (patient satisfaction assessed with OHIP-14 in Esposito 2018, and patient preference in Esposito 2017), and cost-

effectiveness (time to rehabilitate the patient in Esposito 2018, and implant placement time in Esposito 2017).

[Table 1](#) shows which studies contributed data to each outcome domain. We did not perform a quantitative meta-analysis because the studies addressed different clinical aims (comparison of reconstructive strategies in Esposito 2018 versus comparison of site-preparation techniques in Esposito 2017). No major assumptions were required for data inclusion, as outcomes were clearly reported at the patient level in the parallel-group trial and at the side level in the split-mouth trial.

Excluded studies

We excluded 17 completed or ongoing studies for the following reasons:

- insufficient number of participants (< 10 per group) to meet the eligibility criteria (seven studies: Blanco-Ruiz 2023 [71]; ChiCTR2300070948 [72]; NCT06129695 [73]; NCT06155799 [74]; NCT06227351 [75]; NCT06925438 [76]; NCT06968780 [77]);
- zygomatic implants were not functionally loaded (six studies: Bhalerao 2023 [78, 79]; CTIR/2022/08/044951 [80]; CTIR/2022/10/046495 [81]; Fernández-Ruiz 2021 [82]; NCT06737289 [83]; Penmetsa 2024 [84]);
- ineligible indication: zygomatic implants were not clinically indicated because participants had sufficient residual bone (two studies: Nagib 2024 [85]; NCT05108324 [86]);
- ineligible study design (one study: Korsch 2020 [87]);
- study withdrawn before completion (one study: NCT01961284 [88]).

See [Supplementary material 3](#) for details.

Ongoing studies

We identified four ongoing studies for the following review comparisons: zygomatic implants versus conventional implants in augmented bone (one study with an estimated recruitment of 180 participants; CTIR/2012/03/002468 [89]); different zygomatic implants and techniques (three studies with an estimated recruitment of 86 participants; ChiCTR2000035402 [90]; ChiCTR2000035516 [91]; NCT07124715 [92]). See [Supplementary material 4](#) for details.

Risk of bias in included studies

The assessment of risk of bias for each outcome is summarised in the corresponding analysis tables (Analysis 1.1, Analysis 1.2, Analysis 1.3, Analysis 1.4, Analysis 1.5, Analysis 2.1, Analysis 2.2,

Analysis 2.3, Analysis 2.4, Analysis 2.5). See [Supplementary material 5](#).

For prosthetic failures, implant failures, and complications, we judged both Esposito 2017 and Esposito 2018 to have an overall low risk of bias. Randomisation procedures and allocation concealment were appropriate in both studies, and no important concerns were identified regarding deviations from intended interventions, missing outcome data, outcome measurement, or selective reporting.

For patient satisfaction outcomes assessed using OHIP-14 in Esposito 2018, we judged the risk of bias as high because the outcome was assessed only in participants with successful prosthetic rehabilitation, leading to concerns related to missing outcome data. In contrast, patient satisfaction outcomes in Esposito 2017 were judged at low risk of bias.

For cost-effectiveness outcomes, we rated both studies to be at low risk of bias because we judged that outcome measurement methods and reporting were appropriate and consistent with the prespecified analyses.

Synthesis of results

We analysed data from Esposito 2017 and Esposito 2018 at the longest available follow-up. The studies addressed different clinical questions and could not be pooled in a meta-analysis.

Comparison 1. Zygomatic implants versus conventional dental implants in grafted bone

A single RCT was considered (Esposito 2018). See [Summary of findings 1](#).

Critical outcomes

Prosthetic failures

Six prostheses in the augmentation group failed or could not be fitted, compared with two prostheses in the zygomatic group. Zygomatic implants may reduce prosthetic failures compared with conventional dental implants in grafted bone (OR 0.29, 95% CI 0.05 to 1.56; 1 study, 66 participants; low-certainty evidence; Analysis 1.1). We downgraded the certainty of the evidence by two levels for imprecision: the confidence interval was extremely wide and included the possibility of benefit, no effect, and harm.

Implant failures

Nine participants lost 42 implants in the augmentation group compared to three participants who lost six implants in the zygomatic implant group. Zygomatic implants probably reduce implant failures compared with conventional implants in grafted bone (OR 0.23, 95% CI 0.06 to 0.96; 1 study, 61 participants; moderate-certainty evidence; Analysis 1.2). We downgraded the certainty of the evidence by one level for imprecision: the confidence interval was wide and approached the line of no effect.

Complications

Sixteen participants in the group that received conventional dental implants in grafted bone experienced 30 complications, compared with 29 participants who experienced 55 complications in the zygomatic implant group. Zygomatic implants probably increase the risk of complications compared with conventional implants in

grafted bone (OR 4.53, 95% CI 1.47 to 13.98; 1 study, 66 participants; moderate-certainty evidence; Analysis 1.3).

In the zygomatic implants group, all participants treated at the Italian centres (21 participants) experienced transient (from one week to three months) paraesthesia of the infraorbital nerves (21 cases). Overall, 55 complications occurred in 29 participants. Complications included: sinus epithelium perforation (four cases), major facial and lower lip swelling (two cases), peri-implant mucositis (four cases), pain due to implant failure (three cases), sinusitis (eight cases), zygomatic and periorbital infection evolving into a chronic facial fistula (two cases), peri-implant mucosa recession (one case), abutment screw loosening (four cases), fracture of the resin prosthesis lining (one case), fracture of the multi-unit abutment screw (two cases), implant exposure (one case), peri-implantitis (one case), and prosthetic tooth detachment (one case). In particular, sinusitis was very difficult to treat and in some cases could not be resolved without implant removal.

In the group that received conventional dental implants in grafted bone, 30 complications occurred in 16 participants. Complications included sinus epithelium perforation (15 cases), graft fragmentation at implant placement (one case), nasal epithelium perforation (one case), graft infection (two cases), implant exposure (one case), peri-implant bone loss (one case), implant migration into the nasal cavity (one case), peri-implantitis (three cases), and chipping of the prosthesis lining (five cases).

We judged the certainty of the evidence as moderate, downgrading by one level for imprecision because the confidence interval was wide.

Patient satisfaction (reported using OHIP-14)

Patient satisfaction, reported as oral-health-related quality of life measured using the OHIP-14, was characterised by low mean scores in both groups (approximately 4 on a scale from 0 to 56, where lower scores indicate better quality of life). Patient satisfaction may be similar with zygomatic implants compared with conventional implants in grafted bone, although the estimate is very uncertain (MD 0.40, 95% CI -3.05 to 3.85; 1 study, 60 participants; very low-certainty evidence; Analysis 1.4). The certainty of the evidence was very low, downgraded by two levels for imprecision because the confidence interval was extremely wide and included the possibility of benefit and harm, and by one level for risk of bias because only participants with successful prosthetic rehabilitation were included, introducing potential bias due to missing outcome data.

Important outcomes

Cost-effectiveness (time to rehabilitate the patient)

Time to rehabilitate the patient is probably substantially shorter with zygomatic implants than with conventional dental implants in grafted bone (MD -442.98 days, 95% CI -518.64 to -367.32; 1 study, 64 participants; moderate-certainty evidence; Analysis 1.5), indicating a clear advantage of zygomatic implants in terms of time-related cost-effectiveness. Participants treated with zygomatic implants received a functional prosthesis after a mean of 1.34 days, whereas participants treated with conventional dental implants in grafted bone required a mean of 444.32 days to achieve oral rehabilitation. The certainty of the evidence was moderate, downgraded by one level for imprecision due to the small number of participants.

Comparison 2. Zygomatic implants versus subperiosteal implants

We identified no eligible studies.

Comparison 3. Different zygomatic implant techniques for people with severely atrophic maxillae: piezoelectric surgery versus rotary drills

A single RCT was considered (Esposito 2017). See [Summary of findings 2](#). In the piezoelectric surgery group, two of the 20 planned implant-site preparations could not be completed using piezosurgery due to excessive bone hardness, and were therefore carried out using conventional rotary drills. We analysed the data according to the intention-to-treat principle.

Critical outcomes

Prosthetic failures

Since both implant groups were connected under the same provisional prosthesis, prosthetic failures could not be unequivocally attributed to either preparation technique. One implant per group failed in the same participant 15 days after placement, resulting in the loss of one provisional prosthesis (17 participants). There may be little or no difference between piezoelectric surgery and rotary drills in terms of prosthetic failures (OR 1.00, 95% CI 0.02 to 50.40; 1 study, 17 participants; low-certainty evidence; Analysis 2.1). We judged the certainty of the evidence as low, downgrading by two levels for imprecision because the confidence interval was extremely wide and included the possibility of benefit, no effect, and harm.

Implant failures

In 17 participants, one implant failed in each group in the same participant two weeks after placement. The participant presented with swelling, pain, and signs of infection at both implant sites; both implants were mobile and were removed. There may be little or no difference between piezoelectric surgery and rotary drills in terms of implant failures (OR 1.00, 95% CI 0.02 to 50.40; 1 study, 17 participants; low-certainty evidence; Analysis 2.2). The certainty of the evidence was low: we downgraded by two levels for imprecision because the confidence interval was extremely wide and included the possibility of benefit, no effect, and harm.

Complications

Complications may be similar with rotary drills and piezoelectric surgery (OR 1.50, 95% CI 0.25 to 8.99; 1 study, 20 participants; low-certainty evidence; Analysis 2.3), although the estimate is very uncertain. With rotary drills, eight complications occurred in six participants: two inferior lip burns (both healed spontaneously), one case of sinusitis treated medically, one zygoma fracture preventing immediate implant loading, one bilateral infection, two cases of peri-implant mucositis, and one fractured prosthetic screw that was replaced. With piezoelectric surgery, five complications occurred in five participants: one bilateral infection, one implant exposure, one implant extruding through the zygoma (apical protrusion surgically shortened via a blepharoplasty approach), one case of peri-implant mucositis, and one fractured prosthetic screw that was replaced after occlusal adjustment. Postoperative haematomas were reported to be larger at 11 conventionally drilled sites and similar on both sides in nine participants. We judged the certainty of the evidence as low, downgrading by two levels for

imprecision because the confidence interval was extremely wide and included the possibility of benefit, no effect, and harm.

Patient preference

Sixteen participants found both techniques equally acceptable, while four participants preferred piezoelectric surgery. Hence, patient preference may favour piezoelectric surgery (OR 9.00, 95% CI 0.48 to 167.26; 1 study, 20 participants; low-certainty evidence; Analysis 2.4). We judged the certainty of the evidence for patient preference to be low, downgrading by two levels for imprecision: the confidence interval was extremely wide and included both a strong preference for piezoelectric surgery and equal preference for the two approaches, indicating a high degree of uncertainty in the effect estimate.

Important outcomes

Cost-effectiveness (implant placement time in minutes)

Less time was needed to prepare each implant site using rotary drills (14.35 minutes) than piezoelectric surgery (23.5 minutes), corresponding to a mean difference of 9.15 minutes in favour of rotary drills. Therefore, cost-effectiveness probably favoured rotary drills (MD -9.15 minutes, 95% CI -10.41 to -7.89; 1 study, 20 participants; moderate-certainty evidence; Analysis 2.5). We judged the certainty of the evidence for implant placement time to be moderate, downgrading by one level for imprecision due to the small sample size.

Reporting biases

We found no evidence of reporting bias.

DISCUSSION

See [Summary of findings 1](#) and [Summary of findings 2](#).

Summary of main results

This review included two randomised controlled trials (RCTs) with different clinical aims; therefore, quantitative pooling was not appropriate. One parallel-group RCT with 71 participants compared immediately loaded zygomatic implants with bone augmentation and delayed placement of conventional implants in people with severely atrophic maxillae (Esposito 2018), while one split-mouth RCT with 20 participants compared piezoelectric surgery with rotary drills for zygomatic implant site preparation (Esposito 2017).

Moderate-certainty evidence suggested that zygomatic implants may reduce implant failures compared with conventional implants placed in augmented bone. Moderate-certainty evidence also suggested that zygomatic implants probably substantially shorten the time to rehabilitation. However, zygomatic implants probably increased complications (including sinusitis), which were occasionally difficult to manage and in some cases required implant removal. Evidence regarding prosthetic failures and patient satisfaction ranged from low to very low certainty due to imprecision and wide confidence intervals; therefore, the true effects may differ substantially from the observed estimates. Patient satisfaction (indirectly inferred from oral-health-related quality of life scores) appeared to improve with both treatment strategies, with no clear differences between groups (Esposito 2018).

In the comparison between surgical techniques (Esposito 2017), low-certainty evidence suggested little or no difference between piezoelectric surgery and rotary drills for prosthetic failures, implant failures, complications, and patient preference. Moderate-certainty evidence suggested that rotary drills reduced implant placement time compared with piezoelectric surgery, although some findings, such as the larger postoperative haematomas observed with rotary drills, suggested that piezoelectric surgery may be less traumatic. However, two of the 20 implant-site preparations in the piezosurgery group could not be completed using piezosurgery alone and required the use of rotary drills. The available evidence was limited by the small sample size and wide confidence intervals.

Overall, the certainty of the evidence ranged from low to moderate across outcomes, and the available follow-up was limited to three years. Therefore, larger and longer-term RCTs are needed to better define the balance between the benefits and harms of these interventions.

Limitations of the evidence included in the review

One of the main limitations of the included trials was the relatively small number of participants enrolled (Esposito 2017; Esposito 2018). Consequently, several outcomes were characterised by wide confidence intervals, leading to downgrading of the certainty of the evidence for serious imprecision. In some cases, the confidence intervals included the possibility of benefit, no effect, and harm, resulting in low- or very low-certainty evidence.

Moreover, the relatively short follow-up period (up to three years post-loading for most outcomes) limited the assessment of long-term implant and prosthetic survival. In addition, in the trial by Esposito 2018, only participants with successful prosthetic rehabilitations were included in some analyses of patient-reported outcomes, raising concerns regarding risk of bias due to missing outcome data and contributing to downgrading the evidence certainty.

The available evidence was also limited by methodological and clinical considerations. In the comparison between piezoelectric surgery and rotary drills, two implant-site preparations allocated to piezosurgery required conversion to rotary drills, potentially affecting the interpretation of the findings. Furthermore, the placement of zygomatic implants is a technically demanding surgical procedure requiring experienced operators, and severe complications may occur.

Both trials received industry support, raising concerns regarding potential conflicts of interest despite declarations of independence by the authors. However, in Esposito 2018, the company withdrew its support before the initial data analysis.

Limitations of the review processes

We followed Cochrane methodological standards. Given that the lead author of both included trials is also an author of this review, it might be perceived that there is potential for conflict of interest in study interpretation. However, two independent assessors performed study selection, data extraction, and risk of bias assessments.

In this update, we slightly modified the comparisons considered in the review to better reflect the evolving clinical applications and

technical developments in the use of zygomatic implants over the past 13 years.

Agreements and disagreements with other studies or reviews

Several systematic reviews have attempted to summarise the available evidence on zygomatic implants. However, in the following section, we focus only on those reviews that addressed comparisons similar to those of the present review.

Comparison 1: zygomatic implants versus conventional dental implants placed in grafted bone

To our knowledge, no other systematic review has assessed this comparison.

Comparison 2: zygomatic implants versus subperiosteal implants

One review described as “systematic” has addressed this question; however, it was based on inappropriate study designs, making any tentative conclusions unreliable [36].

Comparison 3: one type of zygomatic implant or surgical technique versus another type or technique

Four systematic reviews compared different surgical techniques for zygomatic implant placement [37, 38, 39, 40]. Kämmerer and colleagues compared the original intra-sinus approach described by Brånemark with a more laterally oriented extra-sinus approach [11, 15, 38]. However, the authors included any clinical study, including uncontrolled designs, in their review, and found no RCTs that investigated this approach. Consequently, the review's conclusions are based on lower-level evidence and should therefore be interpreted with caution.

Fan and colleagues and Traboulsi-Garet and colleagues compared dynamic computer-aided surgery, static computer-aided surgery, and free-hand surgery [37, 40], whereas Sharan and colleagues compared only dynamic computer-aided surgery with free-hand surgery [39]. All three reviews included a wide range of study designs, such as prospective and retrospective observational studies. Fan and colleagues also included six cadaver studies [37], while Sharan and colleagues classified an in vitro study using polyurethane stereolithographic models as an RCT [39]. Although all three reviews included the same RCT [78], we excluded this trial because the zygomatic implants were not loaded.

Because the current review applies substantially different methods, its estimates are not directly comparable with those of previous reviews, and any such comparison might be misleading. This highlights an important limitation of the existing secondary literature: many reviews combine heterogeneous study designs, such as prospective case series, retrospective studies, and longitudinal cohort studies. Although these designs can offer descriptive information on patient selection and surgical approaches, they are at higher risk of bias and cannot determine benefits or harms.

By restricting our review to RCTs, we provide a more rigorous synthesis of the effectiveness of zygomatic implants, although the available evidence remains very limited. Overall, this is the only

systematic review in the literature that considers only RCTs and thus provides higher-quality evidence.

AUTHORS' CONCLUSIONS

Implications for practice

Evidence from a single randomised controlled trial (RCT) with moderate- to low-certainty evidence suggests that, in people with severely atrophic maxillae, zygomatic implants probably reduce implant failures and probably increase complications compared with conventional implants placed in grafted bone. Zygomatic implants may reduce prosthetic failures, although the certainty of this evidence is low and the estimates are imprecise. The effect on patient satisfaction remains very uncertain because the evidence is of low certainty. Time to rehabilitation is probably substantially shorter with zygomatic implants (1.34 versus 444.32 days), based on moderate-certainty evidence from the same trial. However, these findings should be interpreted with caution as the evidence is derived from one small study (71 participants). Although zygomatic implants show promise for the management of severely atrophic maxillae, several unpleasant complications were reported.

Evidence on different surgical techniques for placing zygomatic implants is limited to a single small within-participant trial. Piezoelectric surgery may have little to no effect on prosthetic failures and implant failures compared with rotary drills, and complications may be similar between the two approaches. However, the certainty of this evidence is low because of very serious imprecision. Implant site preparation time is probably slower with piezoelectric surgery, which may not always perforate very dense bone, based on moderate-certainty evidence from one small trial. The combined use of drills and piezosurgery may represent a reasonable approach, but the evidence supporting this is uncertain.

Overall, the currently available evidence for both comparisons is limited in quantity and affected by imprecision, which restricts the confidence with which these findings can be applied in clinical settings.

Implications for research

We need additional adequately powered RCTs with long-term follow-up (at least 10 years) to compare zygomatic implants with bone grafting procedures. Encouragingly, two multicentre RCTs are currently underway comparing zygomatic implants with conventional dental implants placed in augmented bone: one with a target enrolment of 180 participants and another with an estimated enrolment of 20 participants.

There is also a complete absence of randomised evidence comparing zygomatic implants with subperiosteal implants, representing an important gap in the literature.

Further randomised trials are required to establish the optimal surgical approach for zygomatic implant placement. In particular, studies should evaluate the most appropriate positioning of zygomatic implants in relation to the maxillary sinus and the residual alveolar crest.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD004151.pub4](https://doi.org/10.1002/14651858.CD004151.pub4).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Risk of bias

Supplementary material 6 Analyses

Supplementary material 7 Data package

Supplementary material 8 Differences between protocol and review

ADDITIONAL INFORMATION

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- Sign-off Editor (final editorial decision): Fang Hua, Wuhan University School of Stomatology, China;
- Managing Editor (selected peer reviewers, provided editorial guidance to authors, edited the article): Sam Hinsley, Cochrane Central Editorial Service;
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- Peer-reviewers (provided comments and recommended an editorial decision): Qi Yan, School & Hospital of Stomatology, Wuhan University (clinical/content review); Brian Duncan (patient and public review); Nuala Livingstone, Cochrane Evidence Production and Methods Directorate (methods review); Jo Platt, Central Editorial Information Specialist (search review).

Contributions of authors

Co-ordination of review: Marco Esposito (ME)

Content expertise: ME

Identification of studies: Riccardo Federico Visconti (RFV), Ciro Cotticelli (CC), Katia Greco (KG)

Data extraction: RFV, CC, KG
Risk of bias assessment: RFV, CC, KG
Data input and analysis: RFV, Alessandro Ambrosi (AA)
Writing of discussion and conclusions: RFV, ME

Some authors involved in previous published versions of this review in 2003, 2005, and 2013 are no longer included on the author byline: Helen Worthington, Paul Coulthard, and Peter Thomsen. Some of the content retained in this review reflects their contributions.

Declarations of interest

Riccardo Federico Visconti: no interests to declare. Member of the Centro Affiliato Cochrane Italia Università Vita-Salute San Raffaele, Milan, Italy.

Katia Greco: no interests to declare.

Ciro Cotticelli: no interests to declare. Member of the Centro Affiliato Cochrane Italia Università Vita-Salute San Raffaele, Milan, Italy.

Alessandro Ambrosi: no interests to declare. Vice-Director of the Centro Affiliato Cochrane Italia Università Vita-Salute San Raffaele, Milan, Italy.

Marco Esposito is one of the authors of the only two studies included in this review. However, he was not involved in the eligibility assessment, data extraction, risk of bias assessment, or GRADE assessment of those studies. He is also the Director of the Centro Affiliato Cochrane Italia Università Vita-Salute San Raffaele, Milan Italy and Senior Editor of the Cochrane Collaboration. He is a Cochrane editor, although he was not involved in the editorial process for this review.

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What's new

Date	Event	Description
23 July 2026	New search has been performed	Search updated to November 2025.
23 July 2026	New citation required and conclusions have changed	Methods updated. New authorship. New included and excluded studies. Conclusions changed.

Disclaimer: The views and opinions expressed therein are those of the authors and do not necessarily reflect those of either the University of Pennsylvania or The University of Manchester.

- The University of Manchester, UK

Support to Cochrane Oral Health

Registration and protocol

Protocol (2003): DOI: 10.1002/14651858.CD004151 [55]

Original review (2005): DOI: 10.1002/14651858.CD004151.pub2 [51]

Review update (2013): DOI: 10.1002/14651858.CD004151.pub3 [52]

Data, code and other materials

As part of the published Cochrane review, the following are made available for download for users of the Cochrane Library: full search strategies for each database; full citations of each unique report for all studies included, ongoing or awaiting classification, or excluded at the full-text screen, in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data (Supplementary material 7). Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, Review Manager (RevMan).

Disclosure of artificial intelligence use

OpenAI ChatGPT (GPT-5.5; OpenAI, San Francisco, CA, USA) was used between September 2025 and June 2026 solely to support editing, wording refinement, grammar correction, and formatting during the preparation of this review.

No AI system, automation tool, or large language model was used to make, guide, or suggest decisions regarding study eligibility, risk of bias assessment, data extraction, data synthesis, GRADE assessments, statistical analyses, or interpretation of findings.

All AI-generated outputs were critically reviewed, verified, and, where necessary, modified or rejected by the review authors before inclusion in the manuscript. All methodological decisions, extracted data, analyses, and interpretations were independently generated and validated by the authors, who remain fully responsible for the accuracy, rigour, integrity, and final content of the review in accordance with the Responsible use of AI in Evidence Synthesis (RAISE) recommendations.

The use of AI did not replace human oversight at any stage and did not compromise transparency, reproducibility, or methodological standards. The authors declare no conflicts of interest related to the AI tools used in the preparation of this review.

History

Protocol first published: Issue 2, 2003

Review first published: Issue 3, 2003

Date	Event	Description
30 August 2013	New search has been performed	Search updated to June 2013.
30 August 2013	New citation required but conclusions have not changed	Methods updated. New authorship. No new included or excluded studies. Conclusions not changed.
13 June 2008	Amended	Converted to new review format.
17 August 2005	New citation required but conclusions have not changed	Substantive amendment.

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ADDITIONAL TABLES

Table 1. Overview of included studies

Study ID (country)	Study design	Population (N)	Intervention	Comparator	Outcomes reported	Follow-up
Esposito 2018 (Spain, Italy)	Multicentre parallel-group RCT	Adults with severely atrophic edentulous maxilla, without sufficient bone volume for conventional implant placement or allowing only two anterior implants, and residual subantral bone height < 4.0 mm (N = 71)	4 zygomatic implants or 2 zygomatic + 2 anterior conventional implants, (immediately loaded within one week)	Augmentation with a bone substitute (6 months of healing) + 6–8 conventional implants, (delayed loaded at 4 months)	Prosthetic failures, implant failures, complications, patient satisfaction (OHIP-14), cost-effectiveness (time to rehabilitate the patient)	3 years
Esposito 2017 (Italy)	Split-mouth RCT conducted at a single centre by two independent external teams	Adults with severely atrophic edentulous maxilla, precluding conventional implant placement and with residual subantral bone height < 4 mm (N = 20)	Piezoelectric surgery used to prepare two zygomatic implant sites (immediately loaded)	Rotary drills used to prepare two contralateral zygomatic implant sites (immediately loaded)	Prosthetic failures, implant failures, complications, patient preference, cost-effectiveness (implant placement time)	3 years

RCT: randomised controlled trial; **OHIP-14:** Oral Health Impact Profile-14.

